



EXPLORATORY ANALYSIS REPORT

SELF PURSUED,
INDEPENDENT ANALYTIC,
PROJECT,
2026.

EXPLORATORY ANALYSIS INTO CHANGE OF FILM PRODUCTION TRENDS IN BENGALI FILMS INDUSTRY.

COMPARATIVE STUDY OF TWO ERAS IN BENGALI FILMS INDUSTRY, IN
TERMS OF VIEWERSHIP, RATINGS AND GENRE COUNTS.

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India, 2026





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ABOUT THIS REPORT

This project analyzes structural changes in the Bengali film industry using IMDb datasets¹, across two eras: the **classics period (1960s–1990s)** and the **modern period (2000s–2020s)**.

**“THIS PROJECT TESTS WHETHER THE ISSUE IS DECLINING
FILM QUALITY OR A BREAKDOWN IN THE SYSTEM”**

While discussions around the industry often focus on perceived declines in quality, this study instead examines how audience engagement, narrative structure, and visibility mechanisms have evolved over time.

Using metrics such as engagement distributions, runtime variability, genre and the relationship between ratings and audience reach.

The goal is to provide a data-driven view of how cultural attention is distributed—and why certain films achieve recognition while others remain invisible.

¹ All datasets used in this project are sourced from the official IMDb public dataset repository: <https://datasets.imdbws.com/>. These datasets are maintained by IMDb and provide regularly updated metadata, ratings, and localization information for films and television content.



EXECUTIVE SUMMARY

The Bengali film industry has undergone significant structural changes over recent decades. While discussions often frame this shift as a decline in film quality, this analysis examines whether deeper systemic changes in how films are produced, consumed, and amplified may better explain these trends.

KEY FINDING:

**THE MODERN SYSTEM PRODUCES FILMS, BUT NO LONGER RELIABLY
CONVERTS QUALITY INTO CULTURAL VISIBILITY.**

Using IMDb data across the two eras, namely: **the classics period (1960s–1990s)** and **the modern period (2000s–2020s)**, this study evaluates audience engagement patterns, runtime structures, genre composition, and the relationship between perceived quality and audience reach.

The results indicate that while production has diversified and baseline accessibility² has **increased**, the mechanisms that once enabled films to accumulate long-term cultural significance have **weakened**.

² Accessibility increasing has been calculated in this study via the means of increasing audience counts and engagement statistics, this does not include audience demographic information which lies outside the scope of the IMDb dataset.

EXECUTIVE SUMMARY

Key findings from this analysis include,

- Engagement patterns have shifted from concentrated, long-term accumulation to fragmented, short-horizon attention.
- Film runtimes show increased variability, indicating a breakdown of standardized theatrical formats.
- Genre composition remains superficially stable, but shifts in narrative function suggest a move toward faster, consumption-oriented storytelling.
- The relationship between ratings and audience engagement collapses in the modern era, indicating that quality no longer predicts reach.
- Audience attention is highly concentrated, with modern films exhibiting extreme inequality in visibility (Gini = 0.96).

EXECUTIVE SUMMARY

These findings suggest that the observed changes are not driven by a simple decline in creative output, but by a transformation in how attention is distributed within the industry.

While more diverse films are being produced, this diversity³ is not matched by a corresponding diversification in audience engagement.

As a result, the modern ecosystem is characterized by both fragmentation and concentration, many films receive limited attention, while a small number dominate visibility without a consistent link to perceived quality.

Given known limitations in IMDb language tagging and regional attribution, results should be interpreted as indicative patterns rather than precise measurements of the Bengali film ecosystem.

³ Diversity in this context as it interests the study is calculated by the variations or the counts of genres films belong to.

DATA & ANALYTICAL APPROACH

DATA SOURCES

This analysis uses publicly available datasets from IMDb, including title metadata (`title.basics`), audience ratings and vote counts (`title.ratings`), and alternative titles with language information (`title.akas`).

These datasets were merged using the unique title identifier (`tconst`) to create a consolidated dataset for analysis.

DATA PREPARATION

Films were filtered to approximate Bengali-language titles using available language and regional indicators, acknowledging known limitations in IMDb tagging and transliteration.

Titles were grouped into two eras: the classics period (1960s–1990s) and the modern period (2000s–2020s).

To account for heavy-tailed engagement distributions, vote counts were log-transformed (`log_numVotes`).

Basic cleaning steps included removing missing or invalid values in key fields such as runtime, ratings, and vote counts.

DATA & ANALYTICAL APPROACH

ANALYTICAL APPROACH

The analysis is structured around five key questions designed to evaluate structural changes in the industry:

- **Engagement Trends (Q1):** Compared vote distributions and rolling median engagement over time.
- **Runtime Structure (Q2):** Analyzed runtime distributions to assess changes in film format and variability.
- **Genre Composition (Q3):** Evaluated shifts in genre distribution using relative frequency analysis.
- **Ratings vs Engagement (Q4):** Examined the relationship between perceived quality and audience reach using Spearman⁴ rank correlation and trend analysis.
- **Attention Inequality (Q5):** Measured the concentration of audience engagement across films using the Gini coefficient.

These components are designed to collectively assess not only changes in film output, but shifts in how audience attention is distributed and sustained.

⁴ Spearman correlation was used instead of Pearson correlation, to evaluate the relationship between ratings and audience engagement, because the data exhibits heavy-tailed distributions and non-linear patterns

KEY FINDINGS

1. Long-Term Cultural Engagement Trends Have Weakened

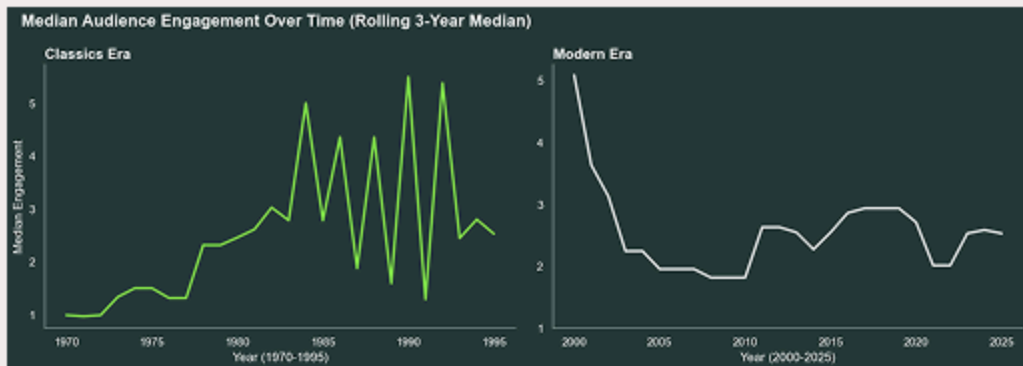


Fig 1: Rolling 3-year median audience engagement trends for the classics era (left) and modern era (right).

The classics era exhibits recurring engagement peaks consistent with films accumulating long-term cultural significance. In contrast, the modern era shows flatter engagement patterns⁵, suggesting that while films continue to receive attention, fewer achieve enduring cultural impact.

⁵ Trends being considered "Flatter" in Modern Era is written in the sense, the modern era graph has relatively lesser quantities of spikes in viewership compared to the classical era, where higher engagement spikes are interpreted as indicators of culturally significant films.

KEY FINDINGS

2. The Long-Tail of Cultural Impact Has Weakened

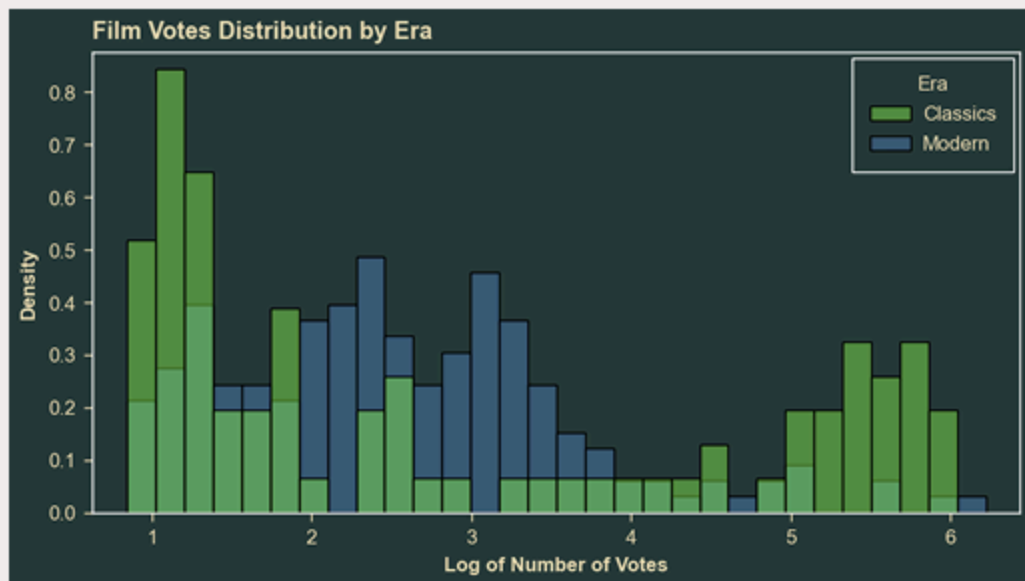


Fig 2: Film vote distributions by era, comparing Classics (green) and Modern (blue) films.

The classics era exhibits a stronger high-engagement tail⁶, consistent with the presence of canonical films accumulating sustained audience attention. In contrast, the modern era shows a more centralized distribution, suggesting that engagement is spread across many titles with fewer enduring cultural anchors.

⁶ The term “high-engagement tail” refers to the presence of films with disproportionately high audience engagement relative to the broader distribution.

KEY FINDINGS

3. Genre Diversity Increased Across the Modern Era

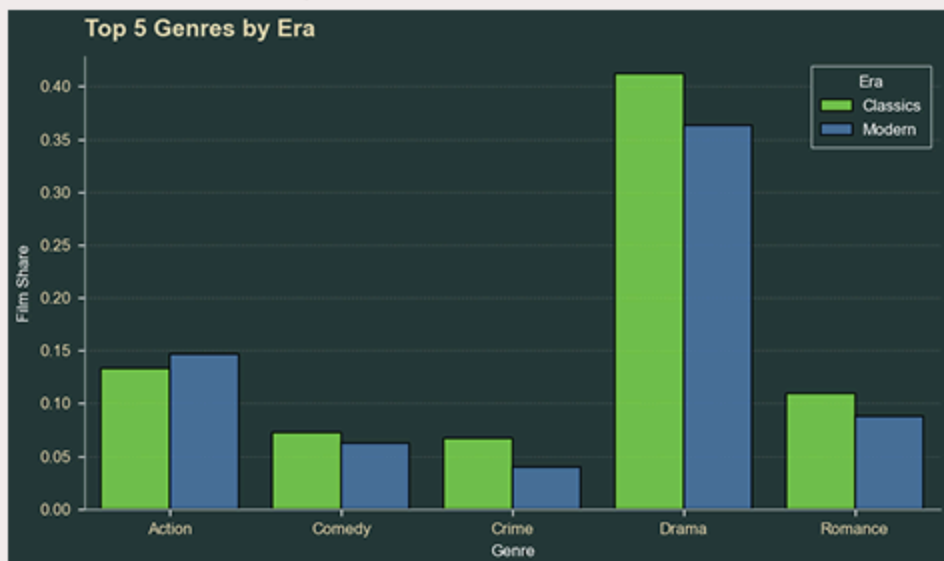


Fig 2: Film vote distributions by era, comparing Classics (green) and Modern (blue) films.

Era	Share of Top 3	Shannon Entropy	Interpretation
Classics	65.45%	2.86	Genre output concentrated around fewer dominant genres
Modern	59.72%	3.24	Genre distribution became more diversified and balanced

Drama remains dominant across both eras, though the modern industry exhibits modest diversification around this stable narrative core.

KEY FINDINGS

4. Quality No Longer Predicts Audience Reach

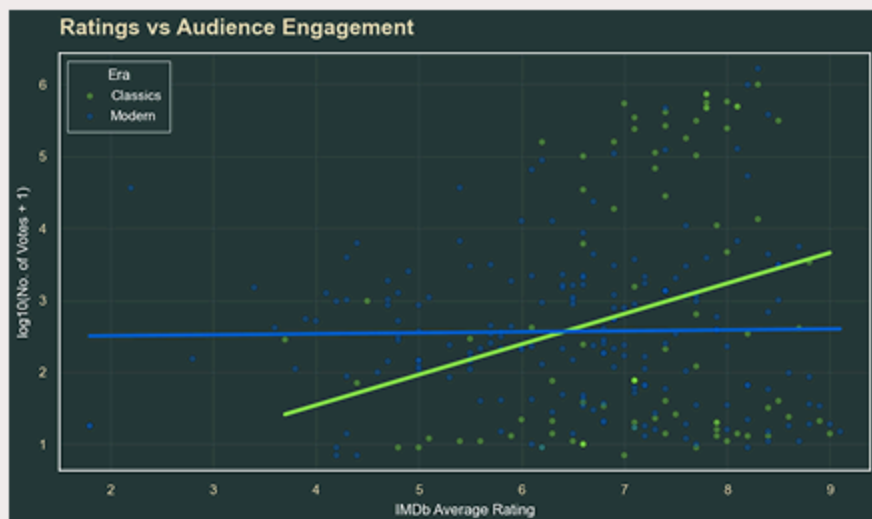


Fig 4: Relationship between IMDb ratings and audience engagement across the classics and modern eras.

Spearman Correlation	Classics	Modern
Spearman $r =$	0.271	-0.048

In the classics era, higher-rated films generally received greater audience engagement, indicating alignment between perceived quality and cultural visibility.

In contrast, this relationship weakens substantially in the modern era, suggesting that audience reach is no longer consistently driven by quality signals.

KEY FINDINGS

5. Audience Attention Has Become Concentrated

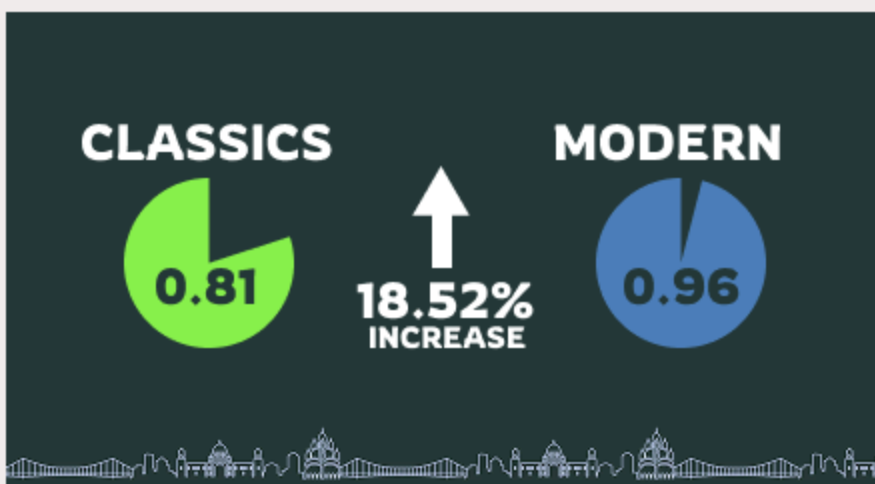


Fig 5: Comparative Gini coefficients measuring audience engagement concentration across eras.

“AN 18.52% INCREASE IN GINI COEFF. SIGNALS ATTENTION BECOMES HYPER- CONCENTRATED”

Audience attention remains unequal across both eras but becomes significantly more concentrated in the modern period. This suggests a shift toward a winner-takes-all visibility structure, where a small number of films capture a disproportionate share of audience visibility.

Despite increased genre diversification, audience engagement itself became less evenly distributed.

SYNTHESIS



Fig 5: Comparative Gini coefficients measuring audience engagement concentration across eras.

“THE MAIN CHANGE WITHIN THE INDUSTRY MAY NOT BE WHAT FILMS ARE BEING MADE, BUT HOW CULTURAL ATTENTION ITSELF IS DISTRIBUTED.”

The findings suggest that the Bengali film industry did not simply undergo a decline, but changed in how audience attention is distributed.

While modern films span a more diverse range of genres and formats, audience engagement has become increasingly concentrated around a smaller number of highly visible films.

At the same time, the relationship between ratings and audience reach has weakened, indicating that perceived quality no longer reliably translates into cultural visibility.

LIMITATIONS

The findings in this report should be interpreted as indicative structural patterns rather than precise measurements of the Bengali film ecosystem.

Key limitations in this analysis include:

- IMDb language and regional tagging systems are imperfect and may exclude or misclassify some Bengali-language films.
- Audience engagement was approximated using IMDb vote counts, which do not fully capture box office performance, television reach, or long-term cultural recall.
- IMDb users represent a digitally active subset of audiences and may not reflect broader demographic viewing behavior.
- The analysis identifies structural relationships and engagement patterns, but does not establish direct causation between variables.

Despite these limitations, the dataset remains useful for identifying broad engagement and visibility trends across eras.

CONCLUSION

This report examined long-term structural changes in the Bengali film industry using IMDb engagement and metadata patterns across the classics and modern eras.

The findings suggest that the industry has not simply undergone a decline in film quality, but a broader transformation in how audience attention is distributed and sustained.

While the modern era exhibits greater diversification in genres and production formats, audience engagement has become increasingly concentrated around a smaller number of highly visible films.

At the same time, the relationship between perceived quality and audience reach has weakened, indicating that visibility no longer consistently aligns with ratings or long-term cultural engagement.

Taken together, the results suggest that the most significant shift within the industry may not be the films being produced, but the structure through which cultural attention itself is accumulated and amplified.

APPENDIX I : DATA PREPROCESSING

Step 1: Dataset merging.

The analysis dataset was constructed by merging multiple publicly available IMDb datasets, including title metadata (title.basics), audience ratings and vote counts (title.ratings), and alternative title/language information (title.akas). Datasets were joined using the unique title identifier (tconst).

Step 2: Dataset filtering.

Films were filtered to approximate Bengali language cinema using available language and regional metadata fields. Due to known limitations in IMDb language tagging and transliteration systems, some titles may have been excluded or misclassified.

Step 3: Dataset Cleaning.

Basic preprocessing steps included the removal of records with missing or invalid values in key analytical fields such as runtime, ratings, and vote counts. Films were subsequently grouped into two analytical eras: the **classics period (1960s–1990s)** and the **modern period (2000s–2020s)**.

APPENDIX I : DATA PREPROCESSING

Step 4: Feature Engineering.

Several derived variables were created to support analysis. Audience engagement was approximated using IMDb vote counts, which were log-transformed (`log_numVotes`) to account for heavy-tailed distributions.

Genre fields containing multiple labels were expanded into individual genre entries for frequency analysis.

Additional derived metrics included rolling median engagement trends, genre share distributions, and Gini coefficients measuring engagement concentration.

“COMPLETE PREPROCESSING SCRIPTS AND ANALYTICAL NOTEBOOKS ARE AVAILABLE IN THE ACCOMPANYING GITHUB REPOSITORY.”

Full pre - processing scripts and analytical notebooks are available in, the accompanying GitHub repository: github.com/ar/repo

APPENDIX II : REPRODUCIBILITY

This project was designed to be reproducible using publicly available IMDb datasets and standard Python-based analytical workflows.

Dataset Access: All datasets used in this analysis are publicly available through the official IMDb datasets repository: <https://datasets.imdbws.com/>

The primary datasets used include:

- `title.basics.tsv.gz`
- `title.ratings.tsv.gz`
- `title.akas.tsv.gz`

Software Environment: The analysis was conducted using Python with standard data analysis and visualization libraries, including:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy

Workflow Summary: The workflow consisted of dataset merging, Bengali-language filtering, preprocessing, feature engineering, exploratory analysis, statistical evaluation, and report visualization.

Repository Access: Complete preprocessing scripts, analytical notebooks, and visualization code are available in the accompanying GitHub repository:
github.com/ar/repo

APPENDIX III : METHODOLOGY

1. Analytical Framework: This report uses a comparative exploratory analysis framework to evaluate structural differences between the classics era (1960s-1990s) and the modern era (2000s-2020s) of Bengali cinema. The analysis focuses on audience engagement, genre composition, runtime structure, and visibility patterns across eras.

2. Exploratory Data Analysis: Distributional differences between eras were evaluated using histograms, boxplots, scatter plots, rolling median trend analysis, and relative frequency comparisons. These visual methods were used to identify long-term shifts in audience engagement and production structure.

3. Statistical Methods: Audience engagement trends were evaluated using rolling 3-year median vote patterns to reduce short-term volatility and highlight sustained engagement behavior over time.

The relationship between IMDb ratings and audience engagement was analyzed using Spearman correlation, selected due to the heavy-tailed and non-linear nature of vote distributions.

Audience attention inequality was measured using the Gini coefficient to evaluate how concentrated engagement remained across films within each era.

4. Genre Analysis Methodology: Genre composition was evaluated by expanding multi-label genre fields into individual genre entries for frequency analysis. Relative frequency comparisons, top-genre share analysis, and Shannon entropy measurements were used to evaluate genre concentration and diversification patterns across eras.

5. Analytical Objective: The primary objective of this analysis was to evaluate whether structural changes within Bengali cinema are associated with increasing audience fragmentation, weakening relationships between quality and reach, and the concentration of cultural visibility around a smaller number of films.
